

UNIT 1	Introduction to Computer Graphics: Overview of Computer Graphics, Computer Graphics Applications, Description of some graphics devices, Input Devices for Operator Interaction, Cathode Ray Tube Basics, Graphic Displays- Random scan displays, Raster scan displays, Frame buffer and video controller, Color CRT, Beam Penetration CRT, The Shadow-Mask CRT, Direct View Storage Tube.
UNIT 2	Scan conversion: Scan-converting a Point, Scan converting a Line, General Requirement for Drawing a Line, DDA Line Algorithm, <u>Bresenham's</u> Line Algorithm, Mid-point criteria, Midpoint Circle Algorithm. Windowing and Clipping: Viewing pipeline, Viewing Transformations, 2-D Clipping algorithms- Line clipping algorithms, Cohen Sutherland line clipping algorithm, Polygon clipping, Sutherland Hodgeman polygon clipping. Polygon Filling, Boundary Fill Algorithm, Flood Fill Algorithm, Scan Line Algorithm
UNIT 3	Two-Dimensional Transformations: Transformations and Matrices, Transformation Conventions, 2D Transformations: Translation, Scaling, Rotation, Reflection, Shearing, Homogeneous Coordinates and Combination of Transformation, Composite Transformations, Transformation of Points, Transformation of Unit Square, Rotation About an Arbitrary Point, Reflection through an Arbitrary Line, The Window-to-Viewport Transformations.
UNIT 4	Three-Dimensional Transformations: Introduction, Three-Dimensional Translation, Three-Dimensional Scaling, Three-Dimensional Shearing, Three-Dimensional Rotation, Three-Dimensional Reflection, Multiple Transformations, Rotation about an Arbitrary Axis in Space, Reflection through an Arbitrary Plane, Matrix Representation of 3D Transformations, Composition of 3D Transformations.
UNIT 5	Visible-Surface Determination: Projection, Parallel Projection, Orthographic Projection, Oblique Projection, Techniques for efficient Visible-Surface Algorithms, Back Face removal, The <u>z-Buffer</u> Algorithm, Scan-line method, Area sub-division method.

AI & ML

Department Elective 2

BCO 019A	ARTIFICIAL INTELLIGENCE	3:0:0
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Course Objective:

- To explain the basic principles of artificial intelligence
- To apply logic and structured concepts in knowledge representation and discuss the applications of artificial intelligence
- To implement and analyze Uninformed and Informed Search Strategies
- To implement and apply various game playing Algorithms to different problems
- Understand and represent various types of logics and their forms
- To Understand and various Learning techniques and analyze concept of ANN

UNIT 1	Introduction- What is intelligence? Foundations of artificial intelligence (AI), Task of artificial intelligence, Techniques of artificial intelligence, Problem Solving Formulating problems, problem types, states and operators, state space. Knowledge Representation- Role of Knowledge, Declarative Knowledge, Procedural Knowledge, Knowledge representation Techniques; conceptual graphs; structured representations; frames, scripts; issues in knowledge representation
UNIT 2	Uninformed & Informed Search Strategies- Breath First Search, Depth First Search, Depth Limited Search, Heuristic Functions, Best First Search, Hill Climbing Algorithm, Problems and solutions of Hill Climbing, Iterative Deepening (IDA), A* algorithm, AO* Algorithm
UNIT 3	Game playing- Introduction, Types of games, Minimax game algorithm, Alpha Beta cut-off procedure , Jug problem, Chess problem, Tiles problem
UNIT 4	Logics- Propositional logics, First Order Predicate Logics (FOPL), Syntax of First Order Predicate Logics, Properties of Wff, Clausal Forms, Conversion to clausal forms
UNIT 5	Learning- Overview of different forms of learning, Supervised base learning, Unsupervised based learning, Introduction to Neural networks:- basic, comparison of human brain and machine, biological neuron, general neuron model, Basic Architecture of Neural Networks, Single Computational Layer: The Perception, Choice of Activation functions, Number of Output Nodes and Loss Functions, applications and advantages of neural networks. Brief introduction to single layer and multiplayer networks